Chapter 4 exercises

DM04-E01

Use induction to prove, for all integers $n \geq 1$,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

DM04-E02

Use strong induction or a least counterexample to prove every integer $n \geq 2$ is a product of primes.

DM04-E03

For $a = -37, d = 8$, find unique integers q, r with $a = qd + r, 0 \leq r < 8$. Explain the quotient and `%` remainder under the truncation-toward-zero rules of C99 and C++11 onward.

DM04-E04

Compute $\gcd(1071, 462)$ by Euclid, then express it as an integer linear combination by extended Euclid.

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DM04-E05

Consider $P(n)$, that the first n positive integers sum to $n(n+1)/2$. A proof omits the base but derives $P(k+1)$ from $P(k)$. Does it prove the claim for all $n \geq 1$? Identify the defect.

DM04-E06

Let $m \geq 1$ be an integer and $a_1, \ldots, a_m \in \mathbb{Z}$. Prove that prime $p \mid a_1 \cdots a_m$ divides at least one factor. Identify the induction hypothesis.

DM04-E07

Prove, for all integers $n \geq 1$,

$$1^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

DM04-E08

Find one integer solution of $1071x + 462y = 42$, then all integer solutions.

DM04-E09

Let $b_0 = 2, b_1 = 5, b_n = 5b_{n-1} - 6b_{n-2}$ for $n \geq 2$. Compute b_2, b_3, b_4, prove $b_n = 3^n + 2^n$ for every integer $n \geq 0$, and state how many base cases your induction proof uses.

DM04-E10

For $N = 2^4 \cdot 3^3 \cdot 5^2 \cdot 7$, count positive divisors, odd divisors, and perfect-square divisors.

DM04-E11

In $\mathbb{Z}[\sqrt{-5}]$, prove multiplicativity of the norm, find all units, and prove three irreducible but not prime. Refute the claim that all nonzero elements of norm below nine are units.

DM04-E12

Treat $\gcd(0, -12), \gcd(-18, 0)$, and $0x + 0y = c$ separately. For integers a, b, c with a, b not both zero and a particular integer solution (x_0, y_0) of $ax + by = c$, derive the general integer solution, including cases $a = 0$ or $b = 0$.

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References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 4.

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